Report on Computational Exercises

A Two-Day Hybrid School on
Molecular and Materials Modeling

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Introduction

This report documents the computational exercises completed during the "Two-Day Hybrid School on Molecular and Materials Modeling" held on August 26–27, 2026 at BLTP JINR, Dubna. All calculations were performed on a personal workstation running Ubuntu Linux with an Intel Core i9-14900KF processor, using open-source software: ASE, Quantum ESPRESSO, MACE, NWChem, MOPAC, xTB, PySCF, and GROMACS.

Part I: Molecular Modeling

Exercise I.1 – Software Verification

Objective: verify that the full software stack (PySCF, MOPAC, xTB, CREST, NWChem, Quantum ESPRESSO, GROMACS, MACE) is installed and functioning correctly.

Results

Software	Version
conda	26.5.3
ASE	3.29.0
MOPAC	23.1.2
NWChem	7.0.2 / 7.2.2
Quantum ESPRESSO	7.5
GROMACS	2026.3
MACE-torch	0.3.16 (PyTorch 2.13.0+cu130, CUDA RTX 3050)
PySCF, xTB, CREST	installed and tested

End-to-end functionality test on H_2O passed 6/6: PySCF E = −2067.6141 eV, MOPAC E = −2.5019 eV, xTB E = −137.9678 eV; NWChem with MPI and pw.x both verified.

Si crystal test via the ASE → QE interface: SCF E = −227.6975 eV (ecutwfc = 20 Ry, 2×2×2 grid, 1.9 s). k-point convergence check: −227.6975 eV (2×2×2) vs −230.0607 eV (4×4×4). 3 out of 3 tests passed.

Exercise I.2 – Performance Benchmarks

Objective: measure the scaling of MOPAC, NWChem and Quantum ESPRESSO with the number of threads/processes.

Results

MOPAC PM7, mini-DNA fragment $C_{40}H_{50}N_{14}O_{26}P_4$ (134 atoms): 1 thread — 27.5 s, 6 threads — 10.8 s (optimum); more threads degrade performance (12 threads — 15.5 s).

NWChem (CH_3• radical, ZORA B3LYP/6-311G): 2 MPI processes — 9.9 s, 4 — 6.8 s, 6 — 6.6 s. Switching to the cc-pVTZ basis increases the runtime from 8.7 to 38.3 s.

Quantum ESPRESSO (Tl_2 SCF with spin-orbit coupling, 50 k-points): near-linear MPI scaling from 1 to 8–24 processes: 45.2 s → 10.1 s (efficiency 0.90–0.99).

MPI processes	1	2	4	8	16	24
Runtime, s	45.2	31.6	20.7	12.1	11.2	10.1

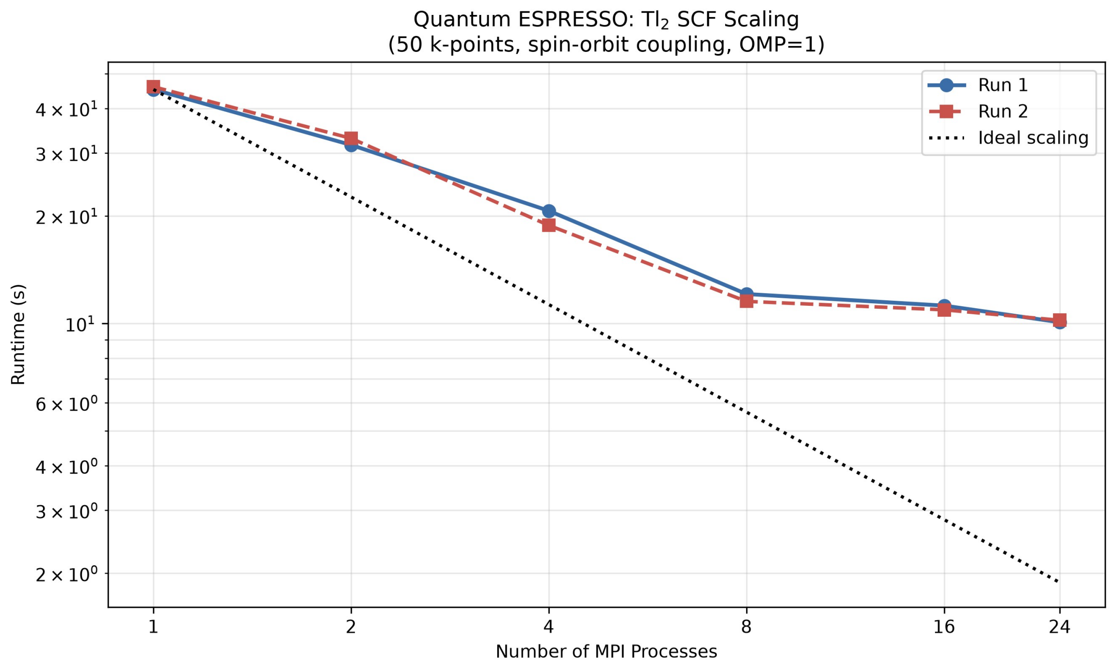

Fig. 1. Quantum ESPRESSO MPI scaling (two runs; dotted line — ideal scaling).

Exercise I.3 – ASE with MACE (Atomization Energies)

Objective: compare the analytical EMT potential and the MACE-MP-0 machine-learned potential on atomization energies of diatomic molecules.

Results

Molecule	EMT, eV	MACE, eV	Experiment, eV	MACE error
H_2	5.3495	4.2065	4.52	6.9 %
N_2	9.9372	10.3130	9.76	5.7 %
O_2	8.5753	5.5264	5.16	7.1 %
F_2	—	−0.5825	1.65	135.3 %

EMT has no parameters for O_2 and F_2 and poorly describes covalent bonds; MACE reproduces the atomization energies of H_2, N_2 and O_2 with 5–7 % error.

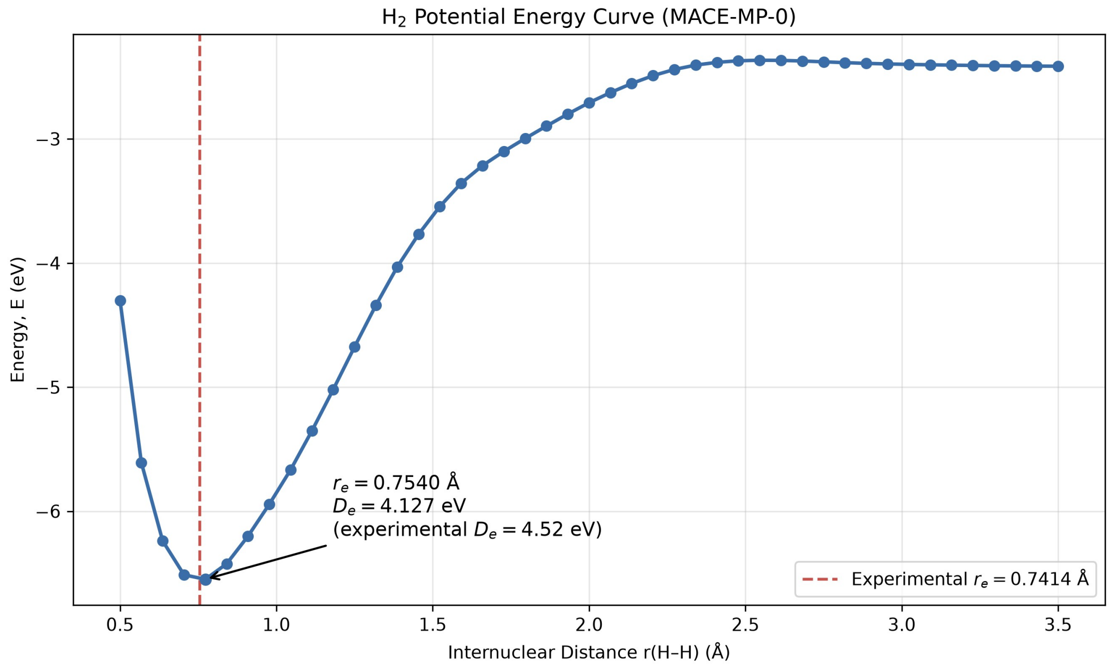

Fig. 2. H_2 potential energy curve (MACE-MP-0): r_e = 0.7540 Å (exp. 0.7414 Å), D_e = 4.127 eV (exp. 4.52 eV).

Exercise I.4 – MACE Installation Tests

Objective: run all MACE installation test scripts independently.

Results

Comprehensive environment check (Python 3.13, PyTorch 2.13.0+cu130, e3nn 0.4.4, CUDA GPU RTX 3050): 22 of 25 checks passed, 0 failed, 3 warnings.

MACE reference energies: H_2O −13.7859 eV, CH_4 −23.1664 eV, NH_3 −18.9956 eV, CO_2 −22.5456 eV, C_6H_6 −75.0697 eV.

Model analysis: mace-osaka26-small (33.5 MB) covers 97 elements (including actinides), MACE-MP-0 — 89 elements; average speed 0.038 s/molecule. MACE water optimization test: E = −14.1603 eV, O–H = 0.9728 Å (4 BFGS steps).

Exercise I.5 – Thermodynamic Properties of HF and O_2

Objective: compute thermodynamic properties with semi-empirical methods (MOPAC PM6/PM7), DFT (NWChem) and a machine-learned potential (MACE), and compare with experiment.

Results (HF)

Property	Experiment	PM7	PM6	B3LYP
r(HF), Å	0.917	—	0.966	—
Frequency, cm^{-1}	4138	3995.8	3968.7	3976.3
ΔfH°(298 K), kJ/mol	−273.3 ± 0.7	−259.1	−266.2	—

Property	Experiment	PM7	PM6	B3LYP
S°(298 K), J/(mol·K)	173.78	173.08	174.33	—
Dipole moment, D	1.827	1.454	1.436	1.859
Polarizability, Å³	0.800	0.815	1.087	0.876

The HF entropy is reproduced within 0.4 %. Additionally (ASE + MACE, O_2, triplet): r(O–O) = 1.2367 Å (exp. 1.2070 Å), frequency 1419.0 cm^{-1} (exp. 1580 cm^{-1}), ZPE 0.0893 eV (exp. 0.0980 eV), S° = 0.002129 eV/K vs experimental 0.002126 eV/K (0.1 % error).

Exercise I.6 – C–C and C–H Bond Energies

Objective: compute the dissociation energies of the C–C bond ($C_2H_6 \rightarrow 2\ CH_3$•) and the C–H bond ($CH_4 \rightarrow CH_3$• + H•) with EMT and MACE.

Results

Property	EMT	MACE	Experiment
d(C–C), Å	1.0096	1.5218	1.5360
E(C–C), eV (kJ/mol)	2.159 (208.3)	3.734 (360.3)	3.81 (368.0)
d(C–H), Å	1.1515	1.0931	1.0870
E(C–H), eV (kJ/mol)	3.254 (314.0)	4.316 (416.4)	4.55 (439.0)

MACE achieves 0.9 % error in bond lengths and 2–5 % in bond energies; EMT is unsuitable (C–C length error 34 %, bond energy error 28–43 %). A separate Methane_CH_Bond_Report.docx was prepared for this exercise.

Exercise I.7 – QE Water Geometry Optimization

Objective: optimize the H_2O geometry in Quantum ESPRESSO (PBE, ecutwfc = 46 Ry) driven from ASE (BFGS).

Results

Six recorded BFGS evaluations (initial + five updates): E = −598.7702 → −598.7717 eV, max force 0.263 → 0.0015 eV/Å. Final: O–H = 0.9734 Å (exp. 0.9578 Å), H–O–H angle = 104.51° (exp. ~104.45°). The bond-length overestimation is typical of PBE.

Challenge I.7 – QE/ASE Methane Relaxation

Objective: adapt the water-relaxation script to CH_4 (PBE, ecutwfc = 46 Ry, Gamma point, 12 Å cubic box).
Results: Total energy = −315.702 eV; mean C–H = 1.0957 Å (exp. 1.087 Å, error 0.8 %); H–C–H = 109.47° (exp. 109.47°, perfect tetrahedral). Max residual force = 0.006 eV/Å. Working files: Challenge_I_7/.

Exercise I.8 – Geometry of $UO_2I_2(OH_2)_2$ (MACE and NWChem)

Objective: optimize the geometry of a U-containing complex — diiododioxouranium(VI) dihydrate $UO_2I_2(OH_2)_2$, 11 atoms (structure from Crawford et al., Inorg. Chem. 2003) — with the MACE machine-learned potential and NWChem DFT.

Results

MACE (mace-osaka26-small, U atom constrained): 15 BFGS steps in 1 s; E = −60.8849 → −61.5744 eV (ΔE = −0.689 eV), fmax 1.241 → 0.011 eV/Å. U=O: 1.7730 → 1.7895 Å (matches the paper within 0.1 Å); U–I: 2.8656 Å (deviates from the paper by −0.16 Å).

NWChem (B3LYP, Stuttgart RLC ECP): optimization converged in 5 steps, E = −376.1745 → −376.2305 a.u. Final bond lengths: U=O 1.7666 Å, U–I 3.0653 Å, U–O(water) 2.4317 Å. A correct description of U–I would require spin-orbit treatment.

Additional Tasks

- Ionization potential of Hg: PM7 — 10.518 eV (exp. 10.437 eV, 0.08 % error); CCSD(T)/def2-QZVP (NWChem) — 10.325 eV; PySCF CCSD(T)/def2-TZVP — 10.125 eV; xTB GFN2 fails (17.66 eV).
- VSEPR: $BeCl_2$, BF_3, PCl_5, SF_6, XeO_4, $Fe(CO)_5$ with MOPAC PM7 and NWChem B3LYP; e.g. d(B–F) in BF_3: PM7 1.305 Å vs experimental 1.307 Å.
- GROMACS 2026.3: crambin 1CRN (46 residues, 642 atoms), CHARMM27/TIP3P: energy minimized to −2582 kJ/mol; NVT equilibration: RMSD 0.139 nm, radius of gyration 0.953 nm, T = 298.1 ± 11.2 K.

Part II: Materials Modeling

Exercise II.1 – Total Energy of Si Unit Cell

Direct QE: E = −227.7156 eV; ASE-driven QE: E = −227.7116 eV. Their 0.0039 eV (0.0017 %) difference confirms the ASE→QE interface; the Fermi level is 6.823 eV.

Challenge

Custom ecutwfc_test.py script tests convergence across 20–80 Ry.

Exercise II.2 – Convergence Testing

Automated sweep of ecutwfc (20–65 Ry) and k-grids (2×2×2 to 14×14×14). Threshold: ΔE < 0.01 meV/atom. A 15×15×15 grid is specified in a separate force/stress input, but its own output file is absent from the repository.

Results

Parameter	Value	ΔE (meV/atom)
ecutwfc	65 Ry	−0.005
K-points	(14, 14, 14)	−0.019

Exercise II.3 – Cell Relaxation of Si

Direct QE vc-relax and ASE BFGS both converge to E = −230.280 eV, a ≈ 5.468 Å. Slightly larger than experimental (5.430 Å at 300 K), consistent with PBE overestimation.

Exercise II.4 – Electronic Properties of Si

SCF → NSCF (30×30×30 k-grid) → DOS/PDOS pipeline. Löwdin analysis, charge density cube, and plots generated.

Results

Atom	Total	s	p
Si	3.964	1.160	2.804

sp^3 hybridization confirmed (s:p ≈ 1:2.4). Spilling: 0.0089 (≪ 0.05). Integrated DOS: 8.0 states (expected: 8 valence electrons). Band gap ~0.6 eV (experimental: 1.12 eV — classic DFT underestimation).

Generated Plots

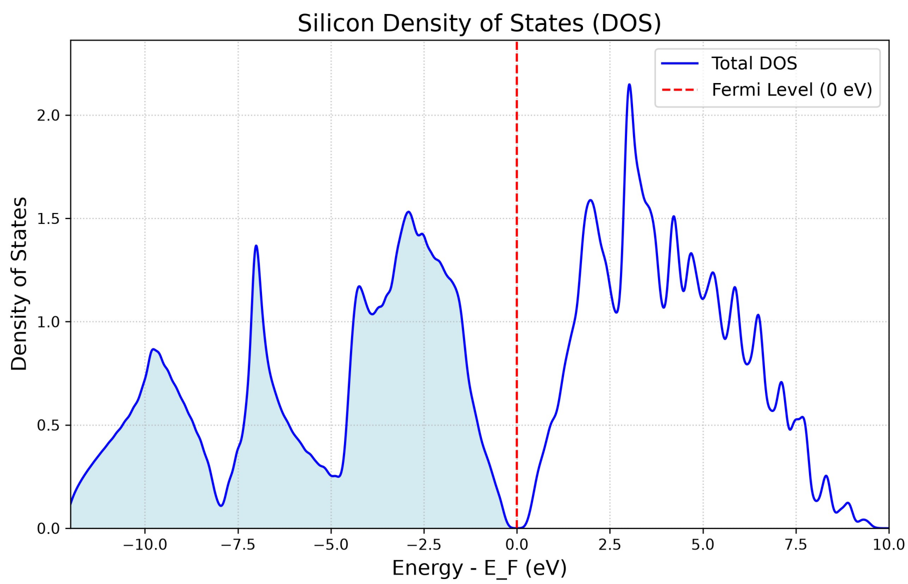

Fig. 3. Silicon Total DOS. Fermi level aligned to zero. Clear band gap visible at E = 0.

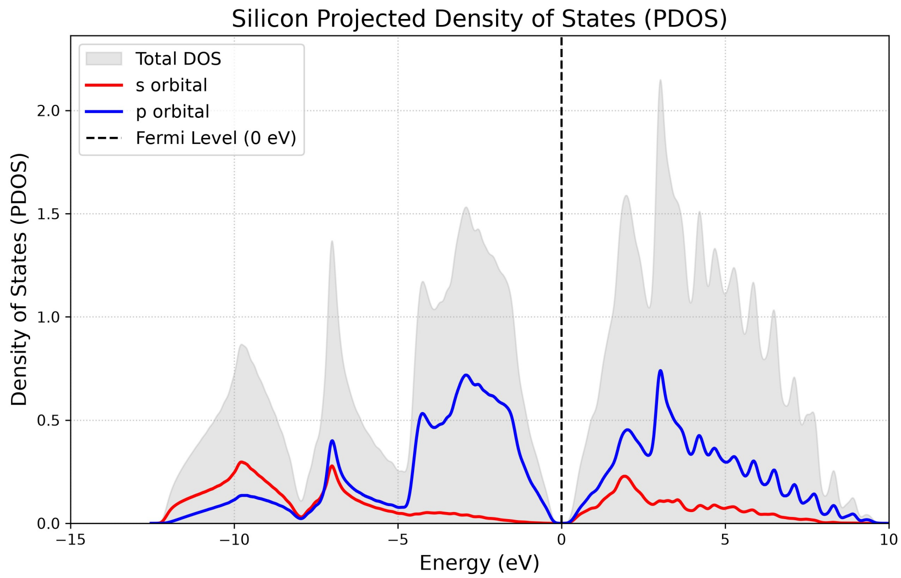

Fig. 4. Silicon PDOS showing s (red) and p (blue) orbital contributions. Mixing in the valence band demonstrates sp³ hybridization.

Challenge II.4 – Effect of Gaussian Broadening (degauss) in Si DOS

Three degauss values tested (0.005, 0.010, 0.020 Ry); all give E = −230.280 eV, confirming energy is insensitive to smearing. DOS at Fermi level: 0.000, 0.000102, 0.012 states/eV — increases with degauss as expected for a semiconductor with a small gap. Working files: Challenge_II_4/.

Exercise II.5 – Electronic Properties of Al (Metal)

E_SCF = −63.067 eV, E_F = 7.969 eV, integrated DOS = 2.997 states (3 valence electrons). Al DOS is continuous through Fermi level — no band gap. This is the defining characteristic of a metal.

Generated Plots

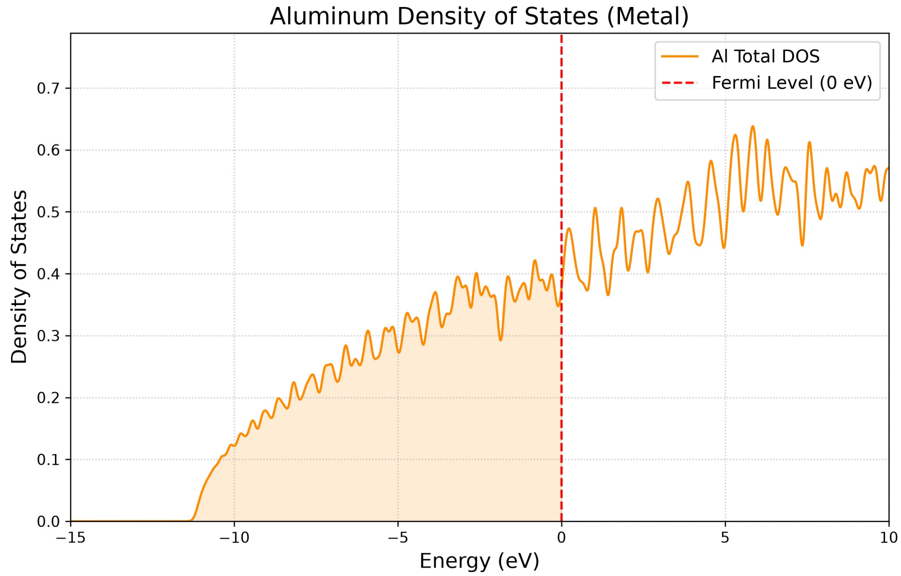

Fig. 5. Aluminum DOS. Continuous spectrum through Fermi level — hallmark of a metal (contrast with Si band gap).

Challenge II.5

Full pipeline in Challenge_Al/: convergence tests, cell relaxation (a = 4.038 Å, primitive-vector components a/2 = 2.019 Å), and electronic properties (24×24×24 NSCF) on relaxed structure. Metal k-point oscillations observed.

Exercise II.6 – Surfaces and 2D Materials: Graphene

Graphene monolayer: 2 C atoms at z = 7.5 Å, c = 15 Å vacuum, assume_isolated = "2D". Converged at ecutwfc = 100 Ry, k-grid (42, 42, 1). Relaxed C–C = 1.424 Å (exp: 1.420 Å). PDOS confirms pz-only Dirac cone at Fermi level.

Generated Plots

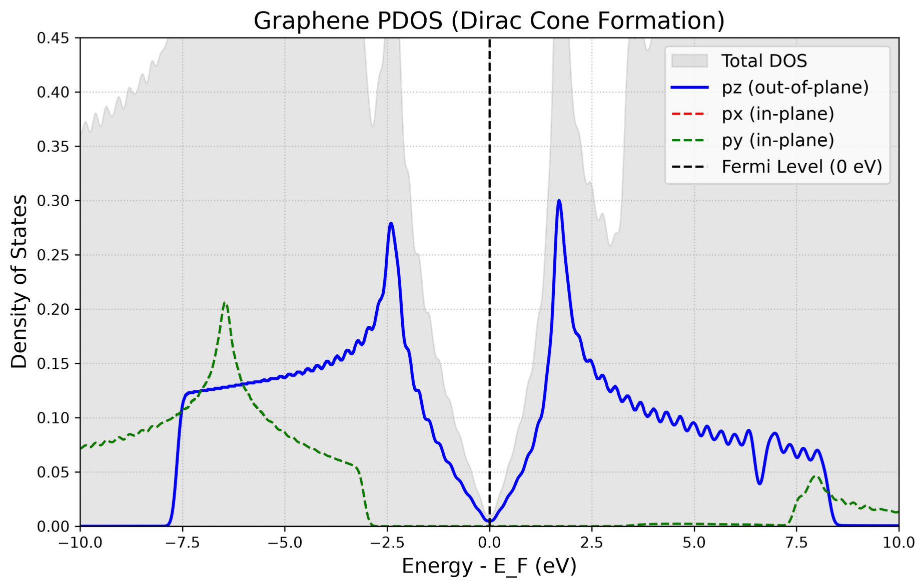

Fig. 6. Graphene PDOS: pz (blue), px (red), py (green). Only pz forms states at Fermi level — the Dirac cone.

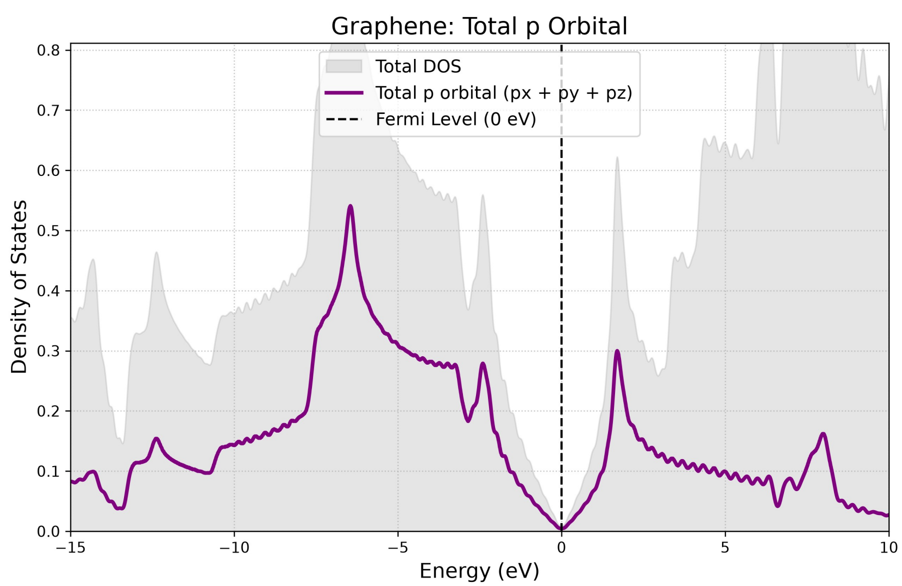

Fig. 7. Graphene total p-orbital. Spectrum converges to zero at Fermi level — semimetallic behavior.

Exercise II.7 – Work Function of Graphene

Work function = minimum energy to extract an electron from graphene surface into vacuum. Computed via SCF → pp.x (plot_num=11) → average.x pipeline.

Results

Property	Value
Fermi Level	−1.717 eV

Property	Value
Vacuum Level	+2.523 eV
Work Function	4.241 eV

Calculated W = 4.24 eV agrees excellently with experimental graphene values (4.2–4.5 eV).

Generated Plots

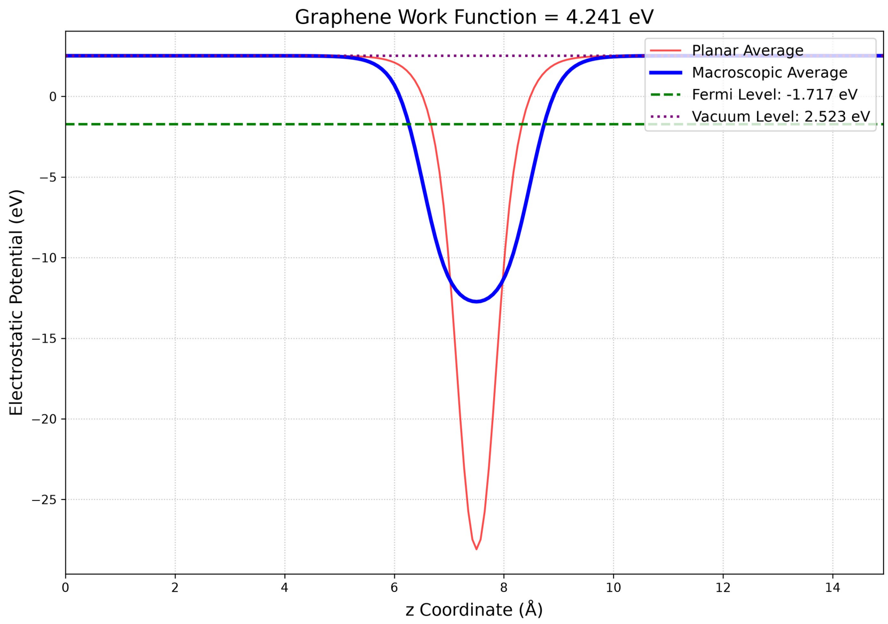

Fig. 8. Electrostatic potential profile along z-axis. Red: planar average, Blue: macroscopic average, Green dashed: Fermi level.

Challenge II.7 – Work Function Without assume_isolated = '2D'

Repeating the graphene work-function pipeline without the 'assume_isolated':'2D' flag removes the artificial potential peaks at cell boundaries.
Results: with 2D: W = 4.241 eV (E_F = −4.241 eV, vacuum = +0.001 eV); without: W = 4.237 eV (E_F = −1.717 eV, vacuum = +2.520 eV). Both agree within 0.004 eV; the potential profile without the flag is physically smoother. Working files: Challenge_II_7/.

Exercise II.8 – Adsorption of H on Graphene

Binding energy of H atom on C_8 graphene flake: E_bind = E(C_8) + E(H) − E(H@C_8).

Results

Quantity	Energy (eV)
E(H@C_8)	−1319.069
E(H)	−12.560
E(C_8)	−1303.646
E_bind	2.864

The C_8 value is elevated by finite-flake edge effects. Extended challenge: Hg on C_{18} was relaxed at top, bridge and hollow sites (E_ads = −0.2378, −0.5294 and −0.6655 eV, respectively); hollow is lowest in energy.

Summary of All Exercises

Exercise	Status	Key Result	Challenge
I.1	√	All packages OK	√
I.2	√	Benchmarked	√
I.3	√	N_2, O_2 atomization	√
I.4	√	MACE OK	22/25
I.5	√	S° matches exp	√
I.6	√	C–H report	√
I.7	√	H_2O optimized	H_2O √; CH_4 √
I.8	√	$UO_2I_2(OH_2)_2$	√
II.1	√	E = −227.71 eV	cutoff √; PP —
II.2	√	65 Ry; sweep to 14^3	base √; extras —
II.3	√	a = 5.468 Å	base √; extras —
II.4	√	Band gap ~0.6 eV	base √; degauss √
II.5	√	Metal DOS	√
II.6	√	Dirac cone pz	√
II.7	√	W = 4.24 eV	base √; rerun √
II.8	√	E_bind = 2.86 eV	Hg sites √; C_8 √

Conclusions

- Mastered complete materials modeling workflow: energy → convergence → relaxation → electronic properties.
- Observed fundamental semiconductor (Si, gap ~0.6 eV) vs metal (Al, continuous DOS) difference.
- Constructed and analyzed 2D graphene: orbital decomposition confirms pz Dirac cone.
- Calculated graphene work function (4.24 eV) from first principles, matching experiment (4.2–4.5 eV).
- Simulated H adsorption on graphene with DFT-D3 van der Waals corrections.
- Assessed DFT limitations: Si band gap underestimation, finite-size effects in adsorption energies.